Week 5 Progress Report
BSE4104 Emerging Trends in Software Engineering
AI-Native & Agentic Engineering
Capstone Project
SME Procurement-Preparation Agent
GROUP J EVENING

1. **Work Completed Against Weekly Objectives**
    - Defined a clear multi-step agent goal.
    - Check inventory - decide if reorder is needed then retrieve quotations and create draft requisition then hand over to humans for approval.
    - Wrote a complete Agent Task Contract covering goal, allowed tools, state, limits, stop conditions and human hand-off rules.
    - Designed and implemented the Sense to Plan to Act to Observe to Stop loop on top of the existing tool-calling orchestrator.
    - Enforced safety limits and strict tool allow list.
    - Updated the system prompt so the agent follows the Task Contract.
    - Captured three execution traces.
    - Updated the architecture diagram to show the bounded agent layer.
    - Committed the Agent Task Contract, updated orchestrator, system prompt and all traces to GitHub.

    The system is now a true bounded agent that it can make multi-step decisions using approved tools and reliably stops when human input is required.

2. **Key Engineering Decisions and Rationale**

Decision	**Rationale**
Single focused goal	Keeps the agent small, testable, and aligned with the “smallest useful system” principle.

Hard tool allow list	Prevents the model from calling non-existent or dangerous functions.
Explicit stop conditions in both code and prompt	Guarantees the agent cannot run indefinitely or take high-impact actions.
Mandatory human hand-off after draft creation	Fully respects the AI Boundary Matrix defined in Week 1.
Three diverse traces	Demonstrates that the agent behaves correctly in normal, edge, and error situations.

3. **Challenges Encountered and Current Response**

Challenges	**Response**
Ensuring the agent stops cleanly after creating a draft	Strengthened the system prompt and added explicit success-stop logic.
Handling invalid item IDs without hallucinating	Agent now correctly surfaces the tool error and requests human assistance.
Keeping the number of steps low while still completing the workflow	Set a practical limit of 8 turns / 6 tool calls; happy-path runs complete in 4–5 steps.
Making traces easy to read for evaluation	Continued using structured JSON logging from Week 4 and added human-readable Markdown summaries.

4. **Repository and Task Management Evidence**
    - GitHub Repository: https://github.com/Nicholas-Muwonge/SME-Procurement-Agent
    - Key additions this week,
        - docs/agent/Agent_Task_Contract_v1.0.md
        - Updated system prompt for bounded agent
        - Enhanced orchestrator with stop-condition handling
        - evidence/traces/TRACE-W5-01
        - evidence/traces/TRACE-W5-02
        - evidence/traces/TRACE-W5-03
    - ClickUp Board: https://app.clickup.com/1200420000004171/v/li/1200420000007247

5. **Plan for Week 6**
   Focus: Memory, State and Interoperability

    - Model workflow/session state explicitly.

- Implement one justified persistent-memory use case
- Document what is stored, why, who can access it, retention and deletion rules.
- Demonstrate that memory improves a legitimate task without silently controlling critical decisions.
- Implement one external integration or document one capability as an MCP-style interface.
- Deliverables: State model, Memory Design Note, working memory demonstration, integration/MCP specification,and Week 6 Progress Report.